1 **(a)** Correct triangle B with vertices at $(-3, 3), (-1, 3), (-1, 6)$ **[B2]**

Translation by vector $\begin{pmatrix} -4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 2 \end{pmatrix}$ **[B1]**

(b) Correct triangle C with vertices at $(-3, 1), (-5, 1), (-5, 4)$ **[B2]**

Reflection in $y = -1$ or correct orientation but wrong position **[B1]**

[Total Marks: 4]

2 **(a)** Correct translation of Shape A **[B2]**

If incorrect, translation of $\begin{pmatrix} -4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 2 \end{pmatrix}$ where $k \neq 0$ **[B1]**

(b) Correct reflection of Shape A to form Shape C with vertices at $(1, -3)$, $(3, -3)$, and $(1, -6)$ **[B2]**

If incorrect, reflection in $x = -1$ or $y = 1$ **[B1]**

[Total Marks: 4]

3 **(a)** Correct triangle with vertices at $(-3, 3), (-1, 3)$ and $(-1, 6)$ **[B2]**

Translation by correct vector but incorrect shape/orientation OR correct size, shape and orientation but wrong translation **[B1]**

(b) Reflection **[B1]**

in the y-axis OR $x = 0$ **[B1]**

[Total Marks: 4]

4 **(a)** Vertices of image plotted correctly at $(-1, -4)$, $(1, -4)$ and $(1, -1)$ **[B2]**

If not B2, award B1 for correct size and orientation, or 2 vertices correct **[B1]**

(b) Reflection **[B1]**

in the y-axis (or $x = 0$) **[B1]**

[Total Marks: 4]

5 **(a)** Vertices of the image at $(3, -2), (1, -2), (2, -1)$ **[B2]**

Correct orientation but wrong position, or translation by $\begin{pmatrix} 4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -3 \end{pmatrix}$ **[B1]**

(b) Vertices of the image at $(1, -1), (1, -3), (2, -2)$ **[B2]**

Reflection in $y = -x$ or reflection in x-axis or y-axis **[B1]**

(c) Vertices of the image at $(-1, 2), (-5, 2), (-3, 4)$ **[B2]**

Correct size and orientation but wrong position, OR enlargement with SF 2 from a different centre **[B1]**

[Total Marks: 6]

6 **(a)** Image B drawn correctly at $(-4, 3), (-2, 3), (-2, 6)$ **[B2]**

If incorrect, 1 mark for translation by $\begin{pmatrix} -5 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 2 \end{pmatrix}$ **[M1]**

(b) Image C drawn correctly at $(1, -1), (3, -1), (3, -4)$ **[B2]**

If incorrect, 1 mark for reflection in y-axis or in a line $y = k$ **[M1]**

(c) Image D drawn correctly at $(-1, -1), (-3, -1), (-3, -4)$ **[B2]**

If incorrect, 1 mark for rotation of 180° about the wrong centre, or 90° about $(0, 0)$ **[M1]**

[Total Marks: 6]

7 **(a)** Translation **[B1]**

By vector $\begin{pmatrix} 6 \\ -4 \end{pmatrix}$ **[B1]**

(b) Correct reflection drawn with vertices at $(-4, -2)$, $(-4, 0)$, and $(-1, 0)$ **[B2]**

Award [B1] if reflected correctly in any horizontal line $y = k$ where $ke1$

(c) Correct rotation drawn with vertices at $(4, 4)$, $(2, 4)$, and $(2, 1)$ **[B3]**

Award [B2] for correct orientation and size in wrong position, or 90° anticlockwise rotation about $(0, 0)$

Award [B1] for any two vertices plotted correctly

[Total Marks: 7]

8 **(a)** Correct shape with vertices at $(-2, -4), (-6, -4), (-6, -6), (-4, -6), (-4, -8)$ and $(-2, -8)$ **[B2]**

Enlargement, scale factor 2 centre $(0, 0)$ OR correct scale factor and orientation but wrong centre **[B1]**

(b) Correct shape with vertices at $(4, -1), (4, -3), (5, -3), (5, -2), (6, -2)$ and $(6, -1)$ **[B2]**

Rotation of 90° clockwise about the wrong centre OR rotation of 90° anti-clockwise about $(1, -1)$ **[B1]**

[Total Marks: 4]

9 **(a)** Rotation **[B1]**

90° clockwise OR 270° anti-clockwise **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Correct triangle drawn with vertices at $(0, 2)$, $(1.5, 2)$, and $(1.5, 3)$ **[B2]**

If incorrect, correct size and orientation but wrong position OR correct centre of enlargement but scale factor 2 **[B1]**

[Total Marks: 5]

10 **(a)** Image B drawn correctly at $(2, -4), (2, -2), (6, -2)$ **[B2]**

If incorrect, 1 mark for reflection in $y = -x$ or correct orientation but wrong position **[M1]**

(b) Image C drawn correctly at $(-4, 2), (-2, 2), (-2, -2)$ **[B2]**

If incorrect, 1 mark for rotation 90° anti-clockwise about $(2, 2)$ or 90° clockwise about the wrong centre **[M1]**

(c) Reflection **[B1]**

in the line $y = 2$ **[B1]**

[Total Marks: 6]

11 **(a)** Vertices of image plotted correctly at $(-1, 2)$, $(-2, 2)$ and $(-2, 0.5)$ **[B2]**

If not B2, award B1 for enlargement SF $-\frac{1}{2}$ from wrong centre, or correct centre but SF $\frac{1}{2}$ **[B1]**

(b) Rotation **[B1]**

180° (direction not required) **[B1]**

centre $(1, 2)$ **[B1]**

(c) Calculates area of $T = \frac{1}{2} \times 2 \times 3 = 3$ **[M1]**

Sets up $3 \times k^2 = 48 \implies k^2 = 16$ **[M1]**

$k = 4$ **[A1]**

[Total Marks: 7]

12 **(a)** Enlargement **[B1]**

Scale factor -2 **[B1]**

Centre $(0, 1)$ **[B1]**

(b) Enlargement scale factor $-\frac{1}{2}$ **[B1]**

Centre $(0, 1)$ **[B1]**

(c) Translating Shape T gives at least two correct coordinates of V: $(-2, 3)$, $(-1, 3)$, $(-1, 5)$ **[M1]**

Recognising reflection in $y = x$ swaps coordinates (x, y)

(y, x)$[M1]$

Final coordinates: $(3, -2)$, $(3, -1)$, $(5, -1)$ **[A1]**

[Total Marks: 8]

13 **(a)** Enlargement **[B1]**

Scale factor -2 **[B1]**

Centre $(1, 0)$ **[B1]**

(b) **(i)** Intermediate triangle V located with vertices at $(0, 1)$, $(-2, 1)$, and $(0, 2)$ **[M1]**

Correct triangle W drawn with vertices at $(0, 3)$, $(-2, 3)$, and $(0, 2)$ **[A2]**

Award [A1] for any two vertices of W plotted correctly

(ii) Rotation **[B1]**

Through $180°$ **[B1]**

About centre $(1, 2)$ **[B1]**

(c) Reflection of (a, b) in $y = x$ gives (b, a) **[B1]**

$(b + 2, a - 4) = (5, -3)$ giving $b = 3$ and $a = 1$ **[M1]**

$(1, 3)$ **[A1]**

[Total Marks: 12]

14 **(a)** Correct triangle drawn with vertices at $(-5, -1)$, $(-2, -1)$, $(-2, 1)$ **[B2]**

If not B2, B1 for correct size and orientation but wrong position OR two correct vertices **[B1]**

(b) Correct triangle drawn with vertices at $(1, 0)$, $(4, 0)$, $(4, -2)$ **[B2]**

If not B2, B1 for reflection in $x = 1$ OR reflection in $y = k$ where $k \neq 1$ **[B1]**

(c) Correct triangle drawn with vertices at $(-1, -2)$, $(-4, -2)$, $(-4, -4)$ **[B2]**

If not B2, B1 for rotation of $90°$ about $(0, 0)$ OR rotation of $180°$ about a different centre **[B1]**

[Total Marks: 6]

15 **(a)** Rotation **[B1]**

$90°$ anticlockwise OR $270°$ clockwise **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Correct shape R with vertices at $(4, 4), (5, 4), (5, 5), (4.5, 5), (4.5, 4.5), (4, 4.5)$ **[B2]**

Any enlargement scale factor $\frac{1}{2}$ OR correct centre $(6, 6)$ with incorrect scale factor **[B1]**

[Total Marks: 5]

16 **(a)** Vertices of image plotted correctly at $(1, 1)$, $(2, 1)$, $(2, 1.5)$, $(1.5, 1.5)$, $(1.5, 2)$ and $(1, 2)$ **[B2]**

If not B2, award B1 for correct size and orientation, or correct enlargement from the wrong centre **[B1]**

(b) Rotation **[B1]**

$90°$ clockwise (or $-90°$) **[B1]**

centre $(0, 0)$ (or origin) **[B1]**

[Total Marks: 5]

17 **(a)** Enlargement **[B1]**

Scale factor $\frac{1}{2}$ (or 0.5) **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Rotation **[B1]**

90° anticlockwise OR 270° clockwise **[B1]**

Centre $(0, 0)$ **[B1]**

(c) Reflection **[B1]**

in the x-axis OR $y = 0$ **[B1]**

[Total Marks: 8]

18 **(a)** Reflection **[B1]**

In the line $y = -x$ **[B1]**

(b) Correct enlargement drawn with vertices at $(0, 0)$, $(2, 0)$, $(2, 1)$, and $(1, 1)$ **[B3]**

Award [B2] for correct enlargement with scale factor $\frac{1}{2}$ from an incorrect centre

Award [B1] for any two vertices plotted correctly

(c) Correct rotation drawn with vertices at $(0, 2)$, $(0, 6)$, $(-2, 6)$, and $(-2, 4)$ **[B3]**

Award [B2] for correct orientation and size in wrong position, or 90° clockwise rotation about $(1, 1)$

Award [B1] for any two vertices plotted correctly

(d) $1 : 4$ **[B1]**

[Total Marks: 9]

19 **(a)** Enlargement **[B1]**

Scale factor -0.5 (or $-\frac{1}{2}$) **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Rotation **[B1]**

90° clockwise (or 270° anti-clockwise) **[B1]**

Centre $(2, 2)$ **[B1]**

(c) Enlargement **[B1]**

Scale factor 0.5 (or $\frac{1}{2}$) **[B1]**

Centre $(0, 0)$ **[B1]**

[Total Marks: 9]

20 **(a)** Enlargement **[B1]**

Centre $(0, 0)$ or origin **[B1]**

Scale factor $-\frac{1}{2}$ **[B1]**

(b) Image R drawn correctly at $(1, 3), (3, 3), (3, 4)$ **[B3]**

If incorrect, 2 marks for correct size and orientation but wrong position **[M2]**

1 mark for correct centre used but wrong scale factor **[M1]**

(c) Rotation **[B1]**

$180°$ **[B1]**

Centre $(0, 1)$ **[B1]**

OR Enlargement [B1], Scale factor -1 [B1], Centre $(0, 1)$ **[B1]**

[Total Marks: 9]

21 (a) Enlargement **[B1]**

Scale factor $-\frac{1}{2}$ **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Reflection **[B1]**

in the x-axis OR $y = 0$ **[B1]**

[Total Marks: 5]

22 (a) Correct triangle U with vertices at $(1, 2), (3, 2)$ and $(3, 0)$ **[B2]**

Enlargement scale factor $\frac{1}{2}$ from wrong centre OR ray lines drawn from $(0, 4)$ but scale factor incorrect **[B1]**

(b) Correct triangle V with vertices at $(2, 1), (2, 3)$ and $(0, 3)$ **[B2]**

Reflection of U in the line $y = -x$ OR reflection of T in the line $y = x$ **[B1]**

(c) Enlargement **[B1]**

Scale factor 2 **[B1]**

Centre $(0, 4)$ **[B1]**

[Total Marks: 7]

23 (a) Correct triangle drawn with vertices at $(-1, -2)$, $(-7, -2)$, $(-7, -6)$ **[B3]**

If not B3, B2 for a correct enlargement of SF -2 but wrong centre **[B2]**

If not B2, B1 for enlargement of SF 2 from centre $(1, 0)$ OR two correct vertices **[B1]**

(b) $(-1, -2)$ **[B1]**

$(-1, -5)$ and $(-3, -5)$ **[B1]**

(c) Rotation **[B1]**

90° clockwise OR 270° anticlockwise **[B1]**

Centre $(7, 3)$ **[B1]**

[Total Marks: 8]

24 **(a)** Subtracts $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ from at least one vertex of U **[M1]**

$(2, 4), (5, 4), (4, 8)$ **[A1]**

(b) Uses the transformation (x, y)

(y, x) $for at least one coordinate [M1]$

$(2, 5), (2, 8), (6, 7)$ **[A1]**

(c) Identifies $(2, 5)$ maps to itself **[B1]**

Applies rotation (x, y)

(y, -x) $relative to the centre (2,5) [M1]$

$(2, 5), (5, 5), (4, 1)$ **[A1]**

(d) Enlargement **[B1]**

Scale factor -2 **[B1]**

Centre $(0, 0)$ **[B1]**

[Total Marks: 10]